

IMAGE EDITING TOOL



Resize



Rotate



Flip



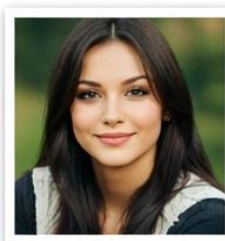
Crop



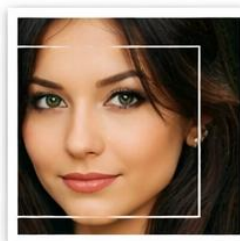
Resize



Rotate



Flip



Crop

Using Python, OpenCV & NumPy



Project Documentation

Name:- Khushi Kumari

Module:- Computer Vision

Q1. A company wants to create a simple image editing tool for their website.

The task:

1. Read and display images
2. Save images
3. Resize, rotate, flip, and crop images
4. Work with image arrays using NumPy
5. Split and merge color channels

You need to implement these features using Python, OpenCV, and NumPy.

Imagine you have a photo on your phone.

This project teaches the computer to:

- Open the photo
- Show the photo
- Change the size of the photo
- Rotate or flip the photo
- Cut a part of the photo
- Change colors in the photo

OpenCV helps us work with images, and NumPy helps us handle image data like numbers in rows and columns.

Project Aim:-

To design and implement a **basic image editing tool** for a company's website that allows users to perform common image operations such as reading, displaying, saving, resizing, rotating, flipping, cropping, and color manipulation.

Requirements:-

- **Programming Language:** Python
- **Libraries:**

- OpenCV (cv2) → for image processing functions
- NumPy → for handling image arrays (pixels as numerical data)

Features to Implement:-

1. Read and Display Images

- Load images using cv2.imread()
- Display images using cv2.imshow()

2. Save Images

- Save edited images using cv2.imwrite()

3. Resize, Rotate, Flip, Crop Images

- Resize → cv2.resize()
- Rotate → cv2.rotate() or transformation matrices
- Flip → cv2.flip()
- Crop → NumPy slicing

4. Work with Image Arrays (NumPy)

- Images are stored as NumPy arrays → rows & columns of pixel values
- Perform mathematical operations on arrays

5. Split and Merge Color Channels

- Split → cv2.split()
- Merge → cv2.merge()

Codes And Screenshot Output:-

```
import cv2
```

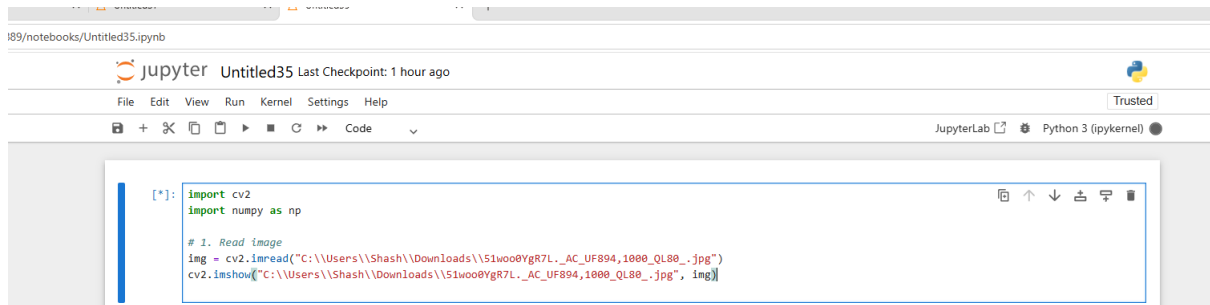
```
import numpy as np
```

```
# 1. Read image
```

```
img =
```

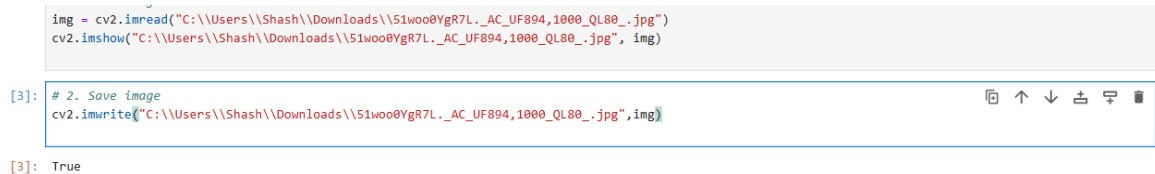
```
cv2.imread("C:\\Users\\Shash\\Downloads\\51woo0YgR7L._AC_UF894,1000_QL80_.jpg")
```

```
cv2.imshow("C:\\Users\\Shash\\Downloads\\51woo0YgR7L._AC_UF894,1000_QL80_.jpg",  
img)
```



2. Save image

`cv2.imwrite("C:\\Users\\Shash\\Downloads\\51woo0YgR7L._AC_UF894,1000_QL80_.jpg",img)`



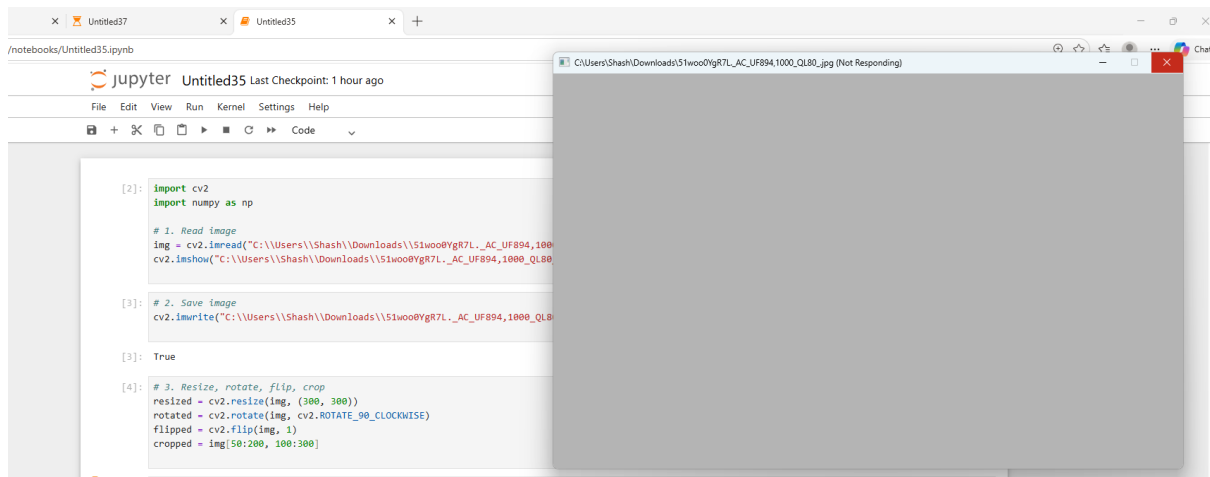
3. Resize, rotate, flip, crop

`resized = cv2.resize(img, (300, 300))`

`rotated = cv2.rotate(img, cv2.ROTATE_90_CLOCKWISE)`

`flipped = cv2.flip(img, 1)`

`cropped = img[50:200, 100:300]`



4. Work with image arrays

`print("Image shape:", img.shape)`

`print("Pixel value at (100,100):", img[100,100])`

```
cropped = img[50:250, 100:300]

[4]: # 4. Work with image arrays
print("Image shape:", img.shape)
print("Pixel value at (100,100):", img[100,100])
|
Image shape: (559, 894, 3)
Pixel value at (100,100): [235 238 243]
```

5. Split and merge channels

```
b, g, r = cv2.split(img)
```

```
merged = cv2.merge([r, g, b])
```

Display results

```
cv2.imshow("Resized", resized)
```

```
[3]: import cv2
import matplotlib.pyplot as plt

resized = cv2.resize(img, (300, 300))

resized_rgb = cv2.cvtColor(resized, cv2.COLOR_BGR2RGB)

plt.imshow(resized_rgb)
plt.axis("on")
plt.show()
```



```
cv2.imshow("Rotated", rotated)
```

```
image shape: (559, 894, 3)  
Pixel Value: [ 0 255  0]
```

```
[9]: # Rotate  
rotated = cv2.rotate(img, cv2.ROTATE_90_CLOCKWISE)  
plt.imshow(cv2.cvtColor(rotated, cv2.COLOR_BGR2RGB))  
plt.axis("off")  
plt.show()
```

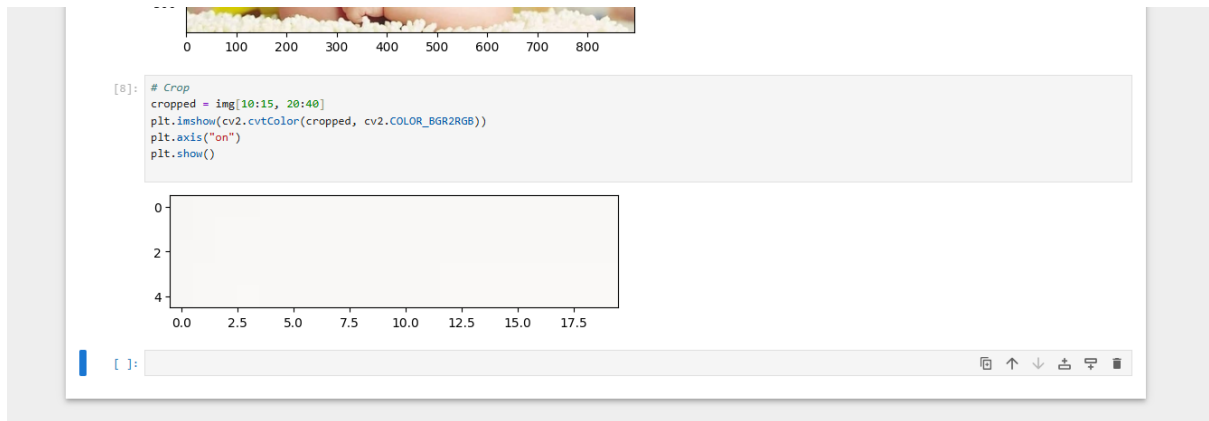


cv2.imshow("Flipped", flipped)

```
[7]: # Flip  
flipped = cv2.flip(img, 1)  
plt.imshow(cv2.cvtColor(flipped, cv2.COLOR_BGR2RGB))  
plt.axis("on")  
plt.show()
```



cv2.imshow("Cropped", cropped)



`cv2.imshow("Merged Channels", merged)`

`cv2.waitKey(0)`

`cv2.destroyAllWindows()`